

The Advance Directive Ontology (ADO)

A Computable Representation of End-of-Life Care Preferences in Advanced Heart Failure

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The problem: directives that fail at the bedside

- **67%** of ICU patients lack decision-making capacity at end of life; fewer than **37%** of US adults have any directive on file.
- The usual story — “directives are unreadable free text” — is *only partly true*: the good forms (California, University of California) are mostly **checkboxes**.

The real problem

Existing instruments are **low-resolution and disease-blind**:

- they collapse *conditional, graded* preferences into a few boxes;
- they ignore the HF-specific decisions — **ICD, LVAD, inotropes**;
- and at 2 a.m. it is **POLST and code-status orders**, not the directive, that get acted on.

ADO — an OWL 2 ontology of EOL preferences:

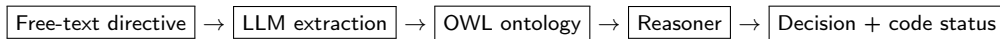
- **67 classes, 22 properties; 5 HF decision points; 21 interventions.**
- Encodes **conditional, graded, negatable** preferences + the patient's **verbatim language**.
- A reasoner returns one of **four honest outputs**.

Our stance

ADO is a **decision-support aid** that helps pre-populate an advance-care-planning note.

It is **not** a legal directive — and complements POLST and the code-status system.

Architecture: ontology *with* LLMs at the boundaries



- The LLM **extracts** (closed-world: encode *only* what the patient stated).
- The ontology is the **auditable, closed-world source of truth**.
- The reasoner produces the **defensible four-category output**.

A pure LLM is non-deterministic and over-confident; an auditable reasoner can honestly say “the directive is silent.”

The honest four-category output

- **Clear** — activation conditions met; preference unambiguous.
- **Partial** — some conditions unmet → defer to surrogate.
- **No coverage** — the directive is silent → escalate, do *not* presume.
- **Vague** — surface the patient's own words (“no heroic measures”).

A checkbox — or a confident LLM — **cannot say “I don’t know.”**
ADO can, and that is the point.

Results I: reasoning + extraction

Scenario-based reasoning

- 16 clinical vignettes, all 5 decision points, all 4 preference types.
- **16/16** decision *and* match-type accuracy.

LLM extraction (real directive text)

- Precision **0.94** / Recall **1.00** / F1 **0.97**.
- Negation, type, conditionality: **15/15**.
- **0** hallucinations on out-of-scope text (nutrition, antibiotics).

The zero-hallucination result is the evidence behind “no coverage.”

Results II (flagship): can ADO produce what governs the bedside?

12 preference profiles from **real templates** (CA AHCD, Texas OOH-DNR, NHS Wales ADRT, Five Wishes, VA, dementia directive). Gold standard = **POLST's own semantics** (independent of ADO).

Field	Agreement
Hospital code status	12/12
POLST Section A	12/12
POLST Section B	11/12
Overall	35/36 (97%)

Cohen's κ = 1.00/1.00/0.88

The value-add

Same directive, two scenarios: **DNR** $\leftrightarrow$ **Full code** as the condition is met or not.

ADO carries the **conditionality** a flat POLST checkbox throws away.

Expert review: Dr. David Magnus (Stanford Biomedical Ethics)

- Reframed the project: *"You can't think of these as AHCDs... This is more like a progress note."*
- On our premise: *"It isn't largely a problem of free text"* — the good forms are checkboxes.
- Deepest critique: *"This system is one-dimensional. Patient preferences are often two or more dimensional"* — what they want *and* who decides.

What we changed in response

Repositioned ADO as an ACP-note aid; **built the code-status / POLST output**; scoped the who-decides axis as the next step.

Limitations & future work

- **No legal force** — ADO is one structured input into a relational, evidence-weighted process, not an oracle.
- **One-dimensional today** — add the **who-decides / surrogate-override** axis (highest priority; ~54% of patients would let family override).
- Add **artificial hydration/nutrition** and the POLST **time-limited trial**.
- Validate against **real EHR code-status data** (MIMIC-IV) — credentialing in progress.

- ADO is the missing **inference layer** above POLST and the code-status system.
- A clinically honest **four-category output** — including the honest non-answers.
- **16/16** reasoning, **F1 0.97** extraction with zero out-of-scope hallucinations, **97%** code-status mapping.
- Ontology **with** LLMs at the boundaries — not ontology *versus* LLM.

Thank you — questions?

https://github.com/patrickwalsh26/bmds210-AD_Ontology